WEEK 3 · LESSON 14 OF 24 · HANDOUT

AI for Research and Learning

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=UG0DP6nVnrc — open the lesson online to watch inside the course.

🎓 **Go deeper:** Teaching CS50 with AI (2025): AI as a personal tutor — https://www.youtube.com/watch?v=6rAWxGAG6EI

Your goal in this lesson: Use AI to learn faster — summarising readings, explaining difficult ideas, making study plans and practice questions — while doing your own thinking and citing real sources.

AI as your personal tutor

AI can explain the same idea ten different ways until one makes sense, at any time of day, without getting tired. Used well, it makes you a **faster and deeper learner**. Used badly, it makes you dependent and leaves gaps in your knowledge.

Six powerful learning uses

1. **Explain simply** — *"Explain supply and demand in simple words with an example from a Kigali market."*
2. **Explain at levels** — *"Explain this at three levels: a 12-year-old, a first-year student and an expert."*
3. **Summarise readings** — *"Summarise this article in 7 bullet points and list 3 questions it leaves unanswered."*
4. **Quiz yourself** — *"Ask me 10 questions on this topic, one at a time. Wait for my answer and tell me if I am right."*
5. **Make a study plan** — *"My exam is in 3 weeks. I can study 2 hours a day. Make a plan covering these topics: [list]."*
6. **Check your understanding** — *"Here is my explanation of photosynthesis. What is missing or wrong?"*

Studying with your own documents

Tools like **NotebookLM** let you upload your own notes, slides and readings and ask questions about them. Answers point back to *your* sources, which makes them easier to check. This is ideal for revision — but only upload material you are allowed to share.

Research: use AI to start, not to finish

- Use AI to **understand a topic** and to suggest **keywords and questions** to search.
- Then find **real sources** in Google Scholar, your university library, official reports and recognised journals.
- **Read the actual sources** before you cite them. Never cite a reference only because AI gave it to you — it may not exist.
- Write in **your own words** and follow your institution's rules on AI use.

Learning, not just answers

The goal of studying is that **you** can do it. Good rules:

- Try first, then use AI to check or explain.
- Ask AI for **hints**, not full solutions, when practising.
- After an explanation, close the chat and explain it back in your own words.

Remember this: Use AI as a patient tutor and research starter, but read real sources, do your own thinking and write in your own words.

Name: _____ Date: _____

Activity

Task 1

Choose a topic you find difficult.

Task 2

Ask AI to explain it at three levels, then ask it to quiz you with 5 questions, one at a time.

Task 3

Ask for 5 keywords to research the topic, and find **two real sources** (a textbook, an official report or a journal article) using them.

Task 4

Write a 150-word summary in your own words that cites your two real sources.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What is the best way to use AI when practising exercises?

   A. Ask for complete solutions immediately
   B. Try first, then ask for hints or explanations
   C. Never use AI for study
   D. Copy answers into your assignment

2. Why are tools that answer from YOUR uploaded notes useful for revision?

   A. They never make mistakes
   B. They are always free
   C. Their answers point back to your own sources, making them easier to check
   D. They write exams for you

3. An AI gives you a journal reference. What should you do before citing it?

   A. Find and read the actual source to confirm it exists and says that
   B. Change the year
   C. Cite it immediately
   D. Ask a friend to cite it

4. Which prompt helps you CHECK your own understanding?

   A. Give me the exam answers.
   B. Write my essay.
   C. Here is my explanation of photosynthesis. What is missing or wrong?
   D. Tell me a joke.

5. What is the right role of AI in academic research?

   A. To finish the research alone
   B. To generate fake citations
   C. To help you start: understand the topic and find keywords and questions
   D. To replace reading

6. After AI explains something, what strengthens your learning most?

   A. Sharing the chat link
   B. Asking for a longer explanation
   C. Closing the chat and explaining it back in your own words
   D. Reading the explanation five more times

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